

Leveraging Spoken Words to Capture Clinical Reasoning: Assessment of Clinical Reasoning based on AI-enabled discourse analysis

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Disclosures



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Objectives

- ▶ Describe limitations of traditional approaches to assessing clinical reasoning in clinical practice
- ▶ Identify potential proxies of clinical reasoning captured in the clinical environment
- ▶ Explain how features of educator-resident dialogue can serve as proxies for real-time clinical reasoning
- ▶ Describe how ambient technology can capture semantic and discourse markers to provide clinical reasoning assessment of individuals and within teams

Clinical Reasoning is a Core Competency Fundamental to Patient Care

Medicine is a science of uncertainty and an art of probability.

- Sir William Osler

Diagnosis, not drugging, is our chief weapon of offence.

- Sir William Osler



Clinical Reasoning is a Core Competency, but Challenging to Define

Clinical Reasoning: Defining It, Teaching It, Assessing It, Studying It

Larry D. Gruppen, PhD

University of Michigan Medical School, Department of Learning Health Sciences, Ann Arbor, Michigan

Perspective

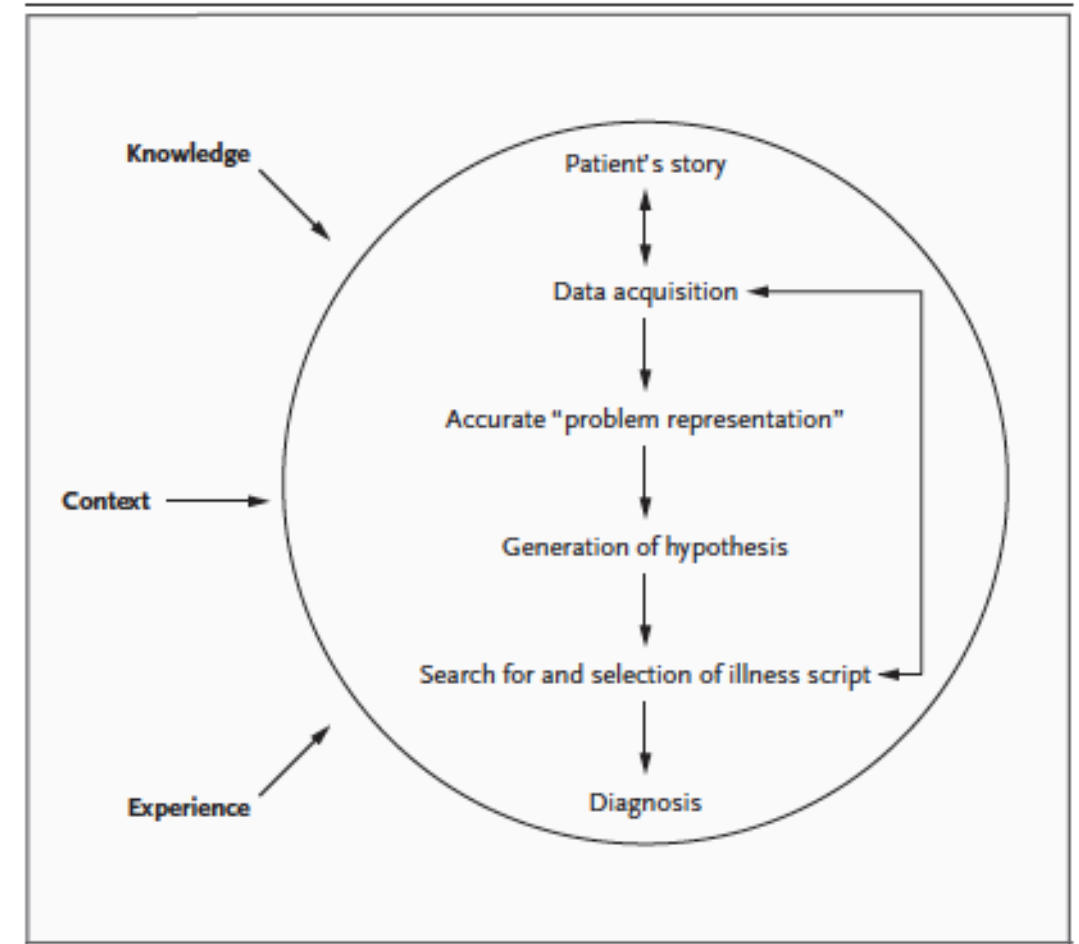
Drawing Boundaries: The Difficulty in Defining Clinical Reasoning

Meredith Young, PhD, Aliko Thomas, PhD, OT, Stuart Lubarsky, MD, MHPE, Tiffany Ballard, MD, David Gordon, MD, Larry D. Gruppen, PhD, Eric Holmboe, MD, Temple Ratcliffe, MD, Joseph Rencic, MD, Lambert Schuwirth, MD, PhD, and Steven J. Durning, MD, PhD

Perspective

Clinical Reasoning as a Core Competency

Denise M. Connor, MD, Steven J. Durning, MD, PhD, and Joseph J. Rencic, MD



Hard to Define, Hard to Assess: The Assessment Landscape of Clinical Reasoning

- ▶ No single tool is sufficient for clinical reasoning measurement
- ▶ Reasoning must be sampled with multiple methods, across multiple contexts, over time
- ▶ Dozens of tools available which can be stratified into:
 - Non-workplace-based assessments and/or simulated assessments (standardized)
 - Workplace-based assessments (authentic)

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- ▶ Dozens of tools available which can be stratified into:
 - Non-workplace-based assessments and/or simulated assessments (standardized)
 - Workplace-based assessments (authentic)
- ▶ Multiple challenges with capturing authentic assessments



**Depends on (limited)
faculty time**

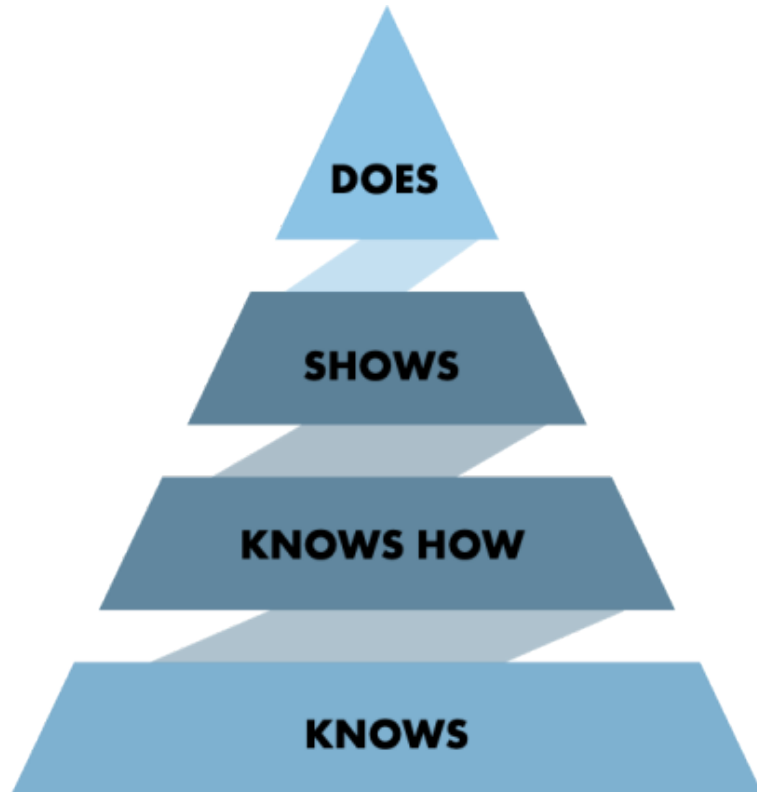


**Captures the product
and not the process**

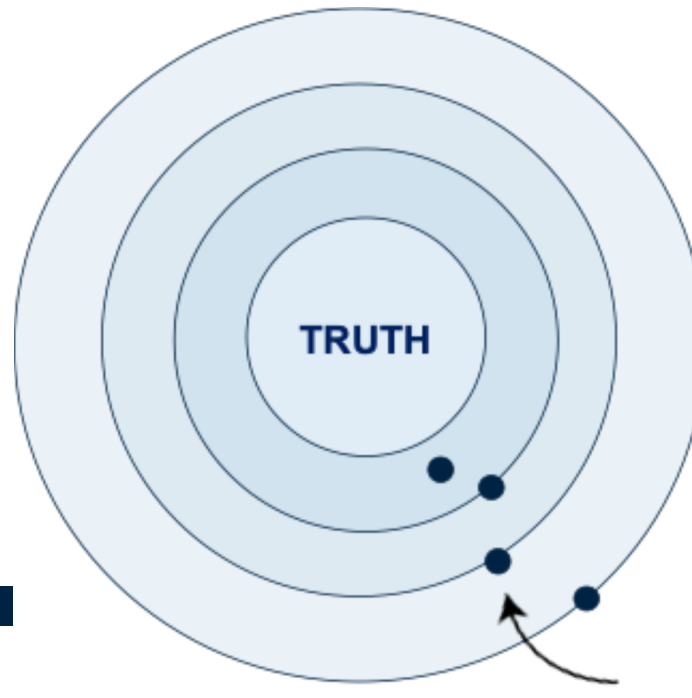


**Ignores the team &
contextual factors**

One's True Clinical Reasoning is Internal / Unknowable... But it Does Leave Traces in the Clinical Environment

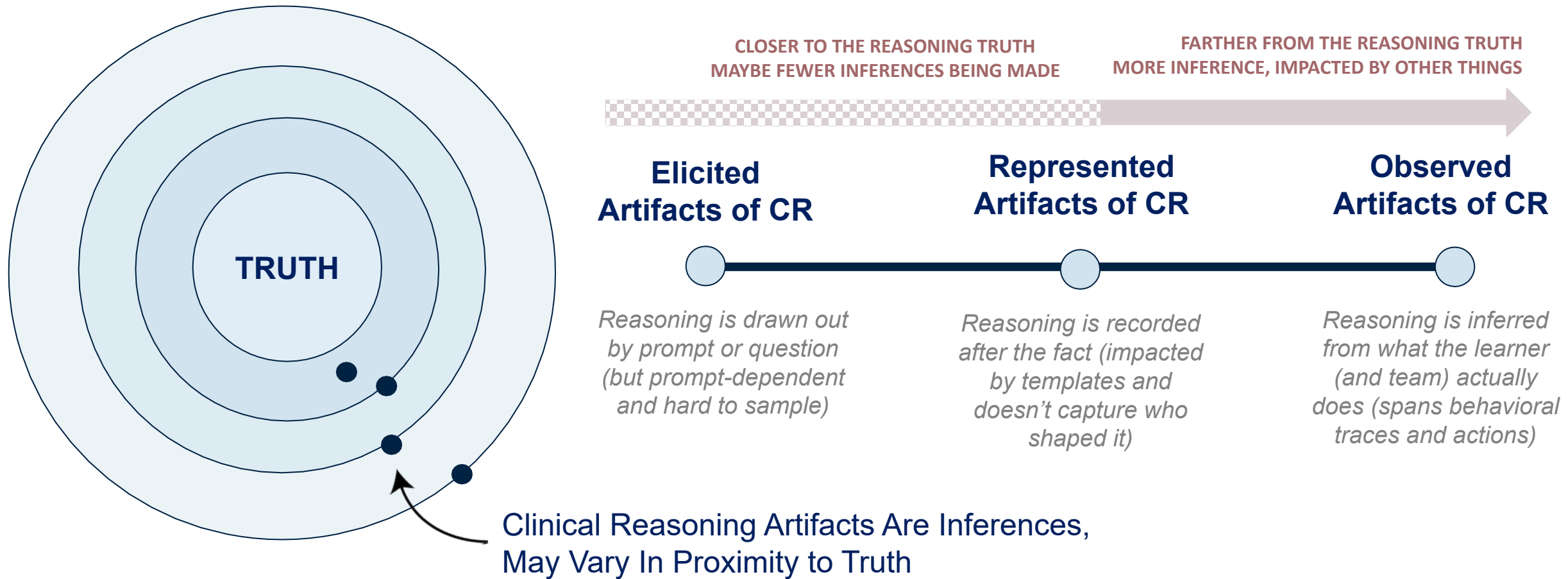


- ▶ Clinical Reasoning Measures that aim to assess what a learner “does” are inferences of the truth
- ▶ Various measures can “artifacts” of reasoning and provide valuable insights into own’s CR



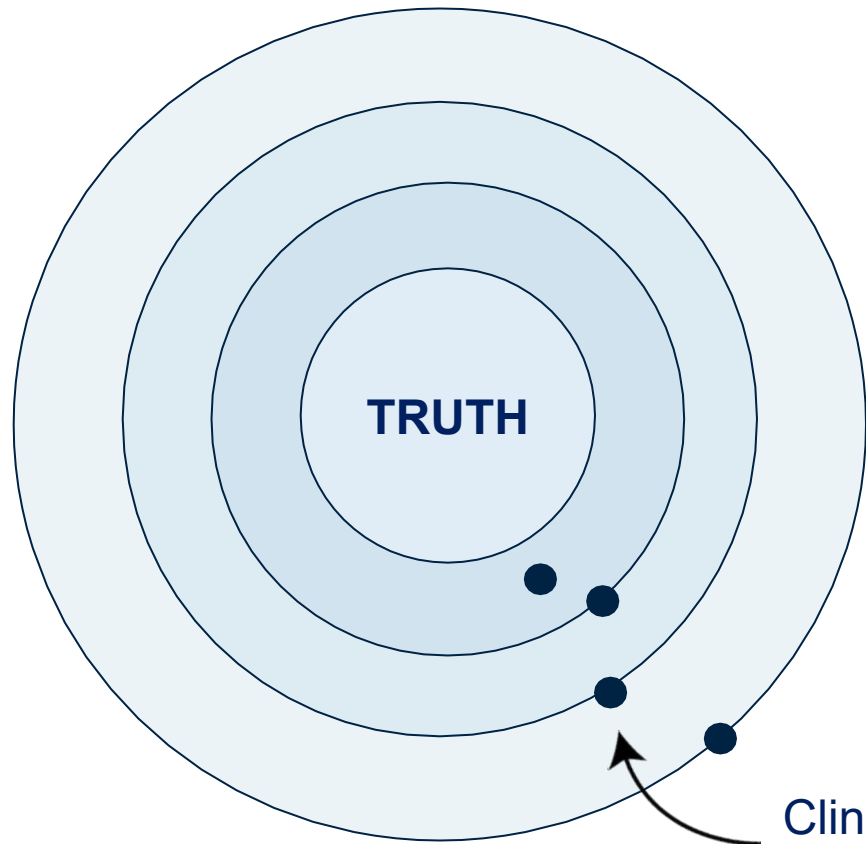
Clinical Reasoning Measures/Artifacts Are Inferences, & May Vary In Proximity to Truth

One's True Clinical Reasoning is Internal / Unknowable... But it Does Leave Traces in the Clinical Environment



One's True Clinical Reasoning is Internal / Unknowable...

But it Does Leave Traces in the Clinical Environment



Elicited Artifacts of CR

- Think aloud
- Chart-stimulated recall
- “What’s your differential”

Represented Artifacts of CR

- Documentation
- Problem lists
- One-liner
- Presentations

Observed Artifacts of CR

- Chart review
- Audit logs
- Gaze tracking
- Direct observation of history taking
- Orders

Clinical Reasoning Artifacts Are Inferences,
May Vary In Proximity to Truth

Three Theoretical Lenses Which Shape Our Understanding of Clinical Reasoning



Script theory

- We use goal-directed knowledge structures adapted to perform tasks efficiently
- Expert reasoning runs on illness scripts
- Novice scripts are sparse / poorly linked
- Expertise shows up in how scripts are activated and compared



Semantic & discourse development *

- Use of semantic qualifiers (acute/chronic, unilateral/bilateral) can indicate expertise
- Discourse patterns can also indicate markers of expertise
 - Reduced
 - Scattered
 - Elaborated
 - Compiled



Situated cognition

- Knowledge, thinking, and learning are situated (or located) in experience, and context is paramount to this
- Reasoning emerges from the interaction of clinician, patient, team, and environment
- Performance varies with context

The *Spoken Word* as a Lens into Clinical Reasoning



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The Spoken Word as a Lens into Clinical Reasoning: Using Both What is Said and How it is Said

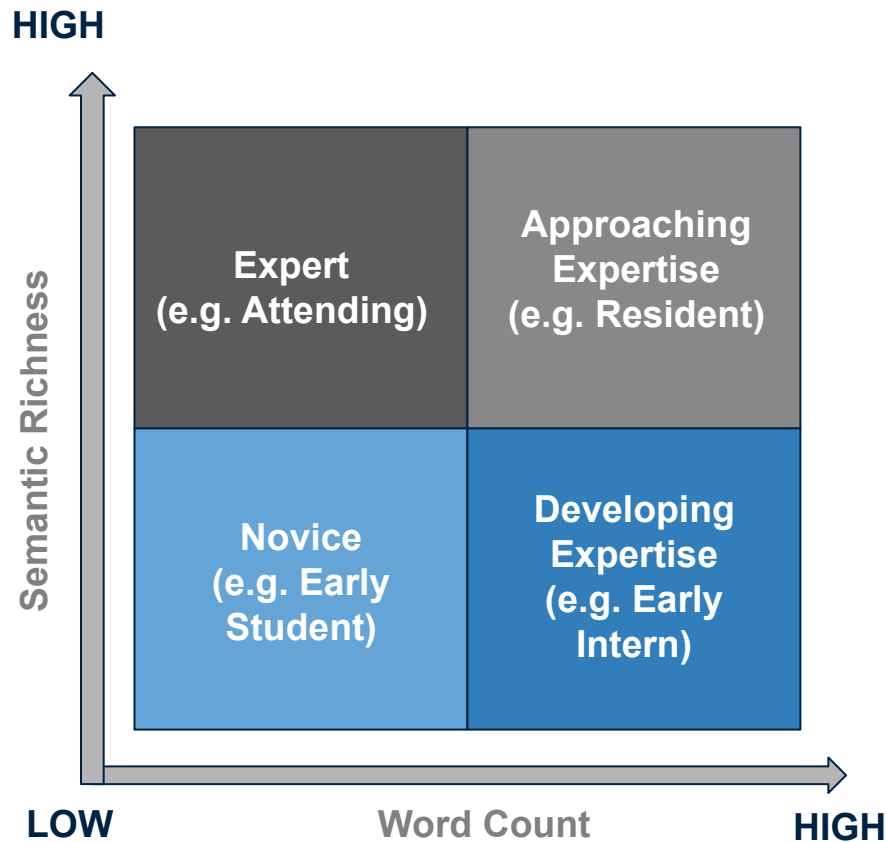
Semantic analysis

- ▶ What was said
- ▶ Presence of semantic qualifiers, differential breadth, linkage language, prioritization language

Discourse analysis

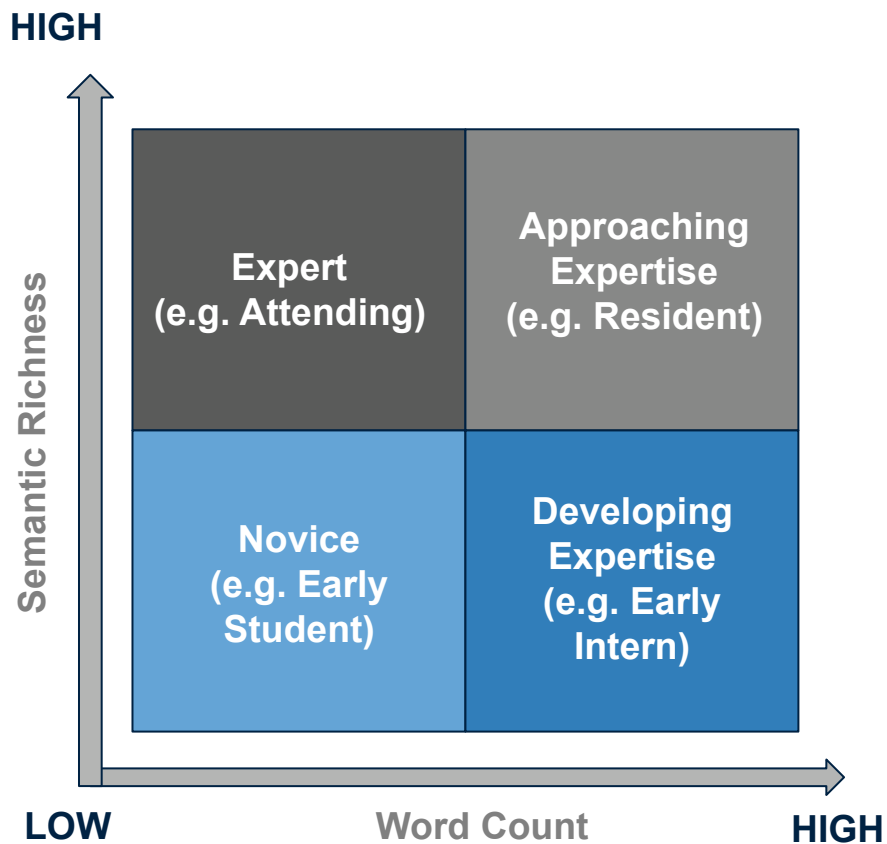
- ▶ How it was said
- ▶ Turn-taking, discourse markers, hedging and certainty, questions versus assertions, interruptions, silences

The Spoken Word as a Lens into Clinical Reasoning: Discourse Provides Developmental Insights for an Individual



Discourse Pattern	Level of Expertise	Example
Reduced	Novice	Few words, low semantic richness, lists findings without qualifiers
Dispersed	Developing Expertise	More words, but content scattered, differential without linkage
Elaborated	Approaching Expertise	Greater semantic richness, qualifiers and linkage between data and diagnosis
Compiled	Expert	Compact, meaning-dense, prioritized, justified, efficient

The Spoken Word as a Lens into Clinical Reasoning: Discourse Provides Developmental Insights for a Team



- Reasoning is co-constructed in a team, demonstrated through hypothesis generation, reframing, questioning, adaptive engagement (highlighting flexibility and judgment)

Discourse Pattern	Level of Expertise	Example
Reduced	Novice	Minimal engagement, deference to authority
Dispersed	Developing Expertise	Asking clarifying questions, seeking validation
Elaborated	Approaching Expertise	Building on others' ideas, introduction of alternatives
Compiled	Expert	Synthesizes team input, inclusive language and framing

Using Discourse for Clinical Reasoning Insights

Resident: I initially thought pneumonia given the cough and recent admission. But she's afebrile, her WBC is normal, and there's no new infiltrate. With the weight gain, edema, and elevated BNP, I think heart failure is more likely.

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Intern: I was worried about COPD since she is wheezing.

Resident: Possibly, although there's no increased sputum. She may have both, but I think heart failure is driving the hypoxemia and perhaps the wheezing.

Attending: What about PE?

Resident: Possible given the recent hospitalization, but less likely right now given the other features. If she doesn't improve with diuresis, I'd reconsider it.

Attending: Bottom line?

Resident: Decompensated HFpEF.

Discourse analysis How is reasoning expressed?	Semantic analysis What reasoning is expressed?
Reasoning initiation	Clinical concepts
Elaboration	Clinical findings
Justification	Relationships between variables
Contrast/comparisons	Directionality
Hedging and uncertainty	Certainty
Revisions	Diagnostic hierarchy
Response to prompts	Causal attribution
Interruptions	Temporal relationships
Change with New Data	Action paired with rationale
Condensing and Synthesis	

Resident: I initially thought pneumonia given the cough and recent admission. But she's afebrile, her WBC is normal, and there's no new infiltrate. With **the weight gain, edema, and elevated BNP, I think heart failure is more likely.**

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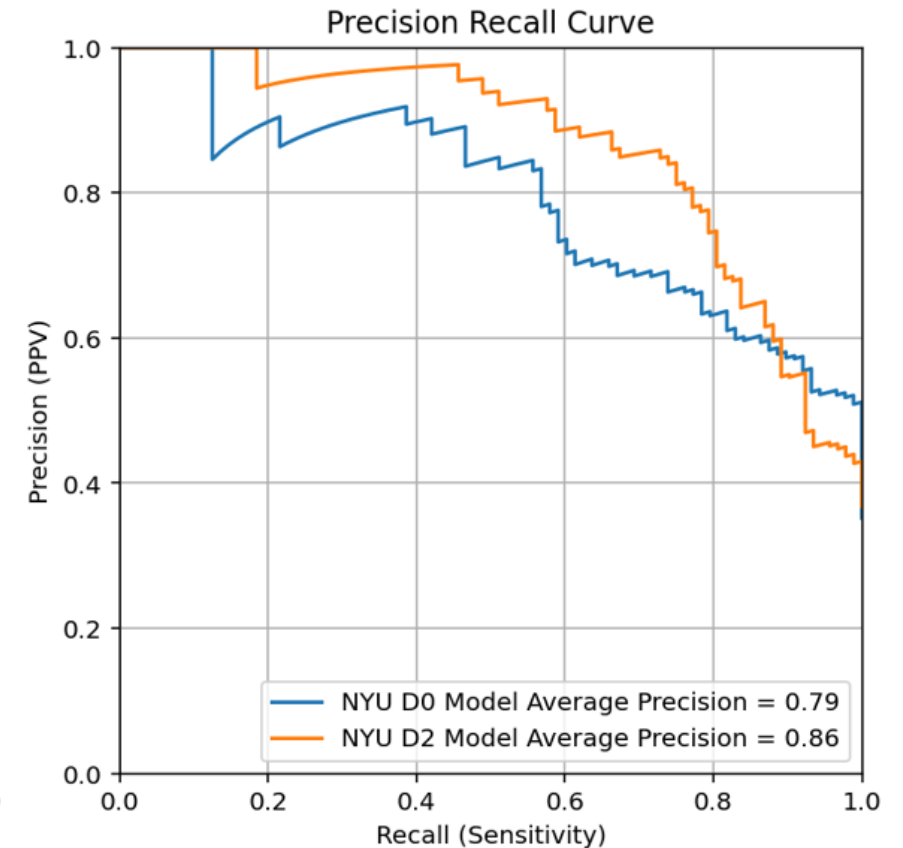
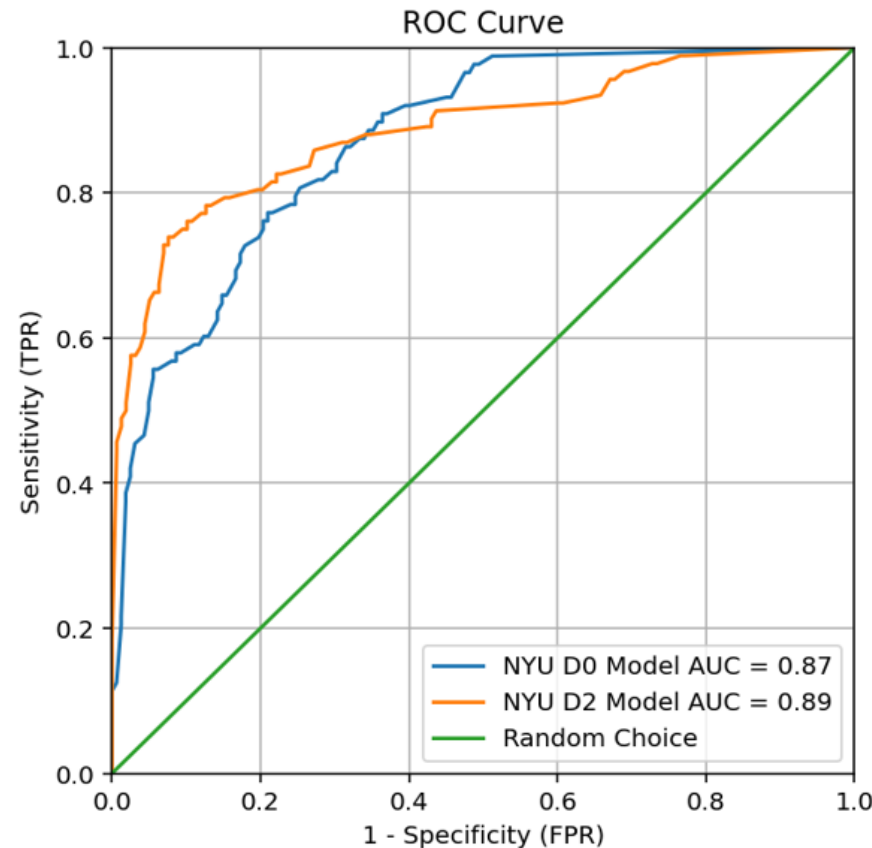
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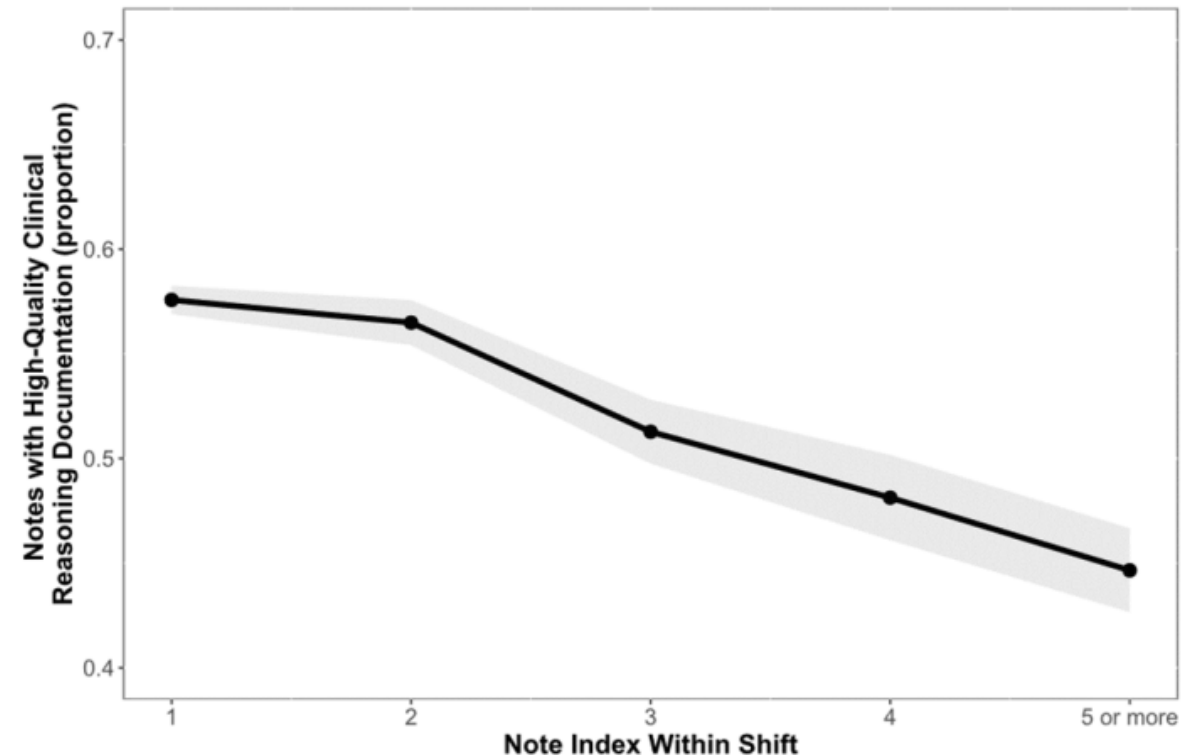
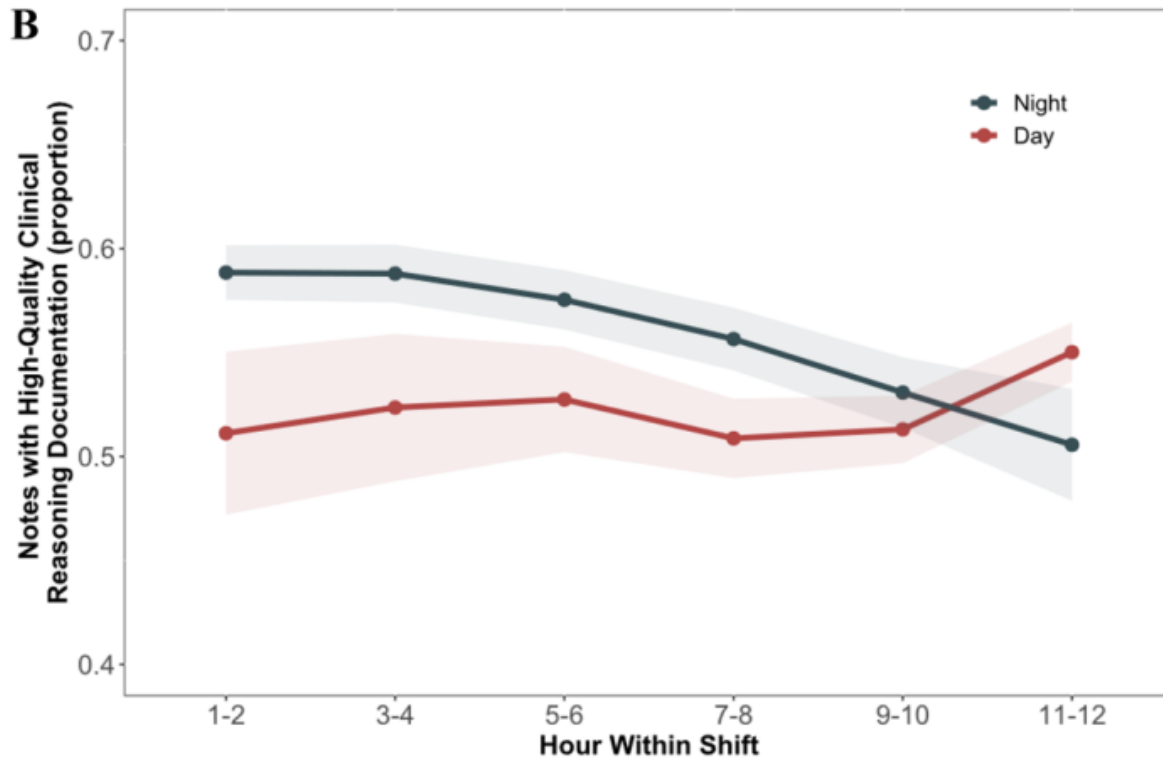
Novel Technologies Now Allow Us to Analyze Clinical Reasoning Artifacts/Traces at Scale

- Schaye, et al. showed potential for analyzing interpretative summaries, differential justification within admission H&Ps (represented artifacts)



Novel Technologies Now Allow Us to Analyze Clinical Reasoning Artifacts/Traces at Scale

- Showed important connection of reasoning proxies to contextual features in clinical environment



Ambient Technology Now Makes Capturing Educator-Learner Discussions Possible as Another Clinical Reasoning Artifact

August 14, 2025 / [Leadership](#) / [Leadership Highlights](#)

Less Typing, More Ambient

Implementa
Adv AI's
time

Original Investi

Use of Ambient AI Burden and Profes

Kristine D. Olson, MD, MSc^{1,2}; Daniell

Ambient Listening: What

Is It and How Is It

Studies find AI technology for clinical
and reduces

- Access to transcripts of conversations opens up new possibilities (source of data which captures elicited and represented reasoning in new ways)
- Advances in technology allows for machine learning to analyze transcripts at scale in a way not previously possible

he Note-Taking Load

documentation burden while
tient conversations,

screen

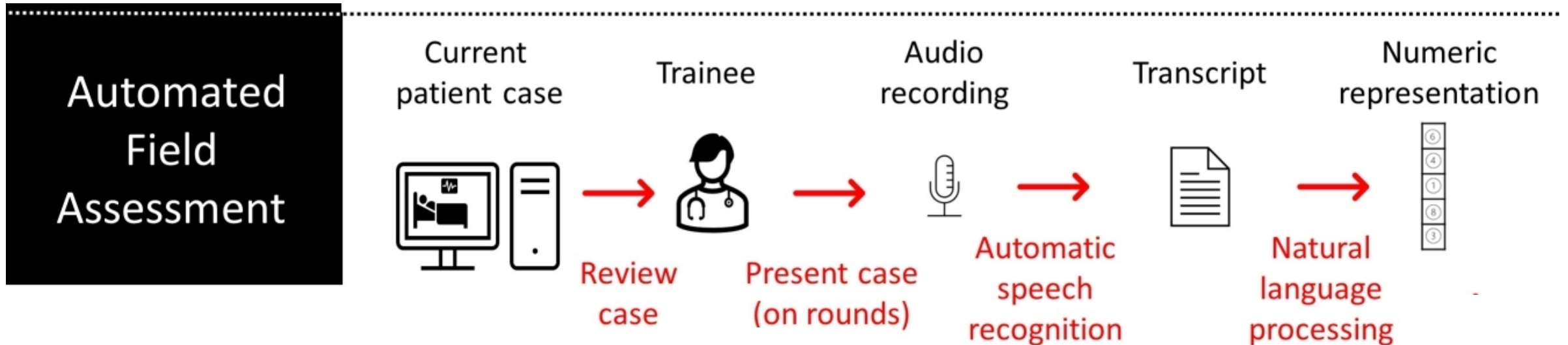
I was a skeptic, but now I'm con

Using technology to reclaim our time

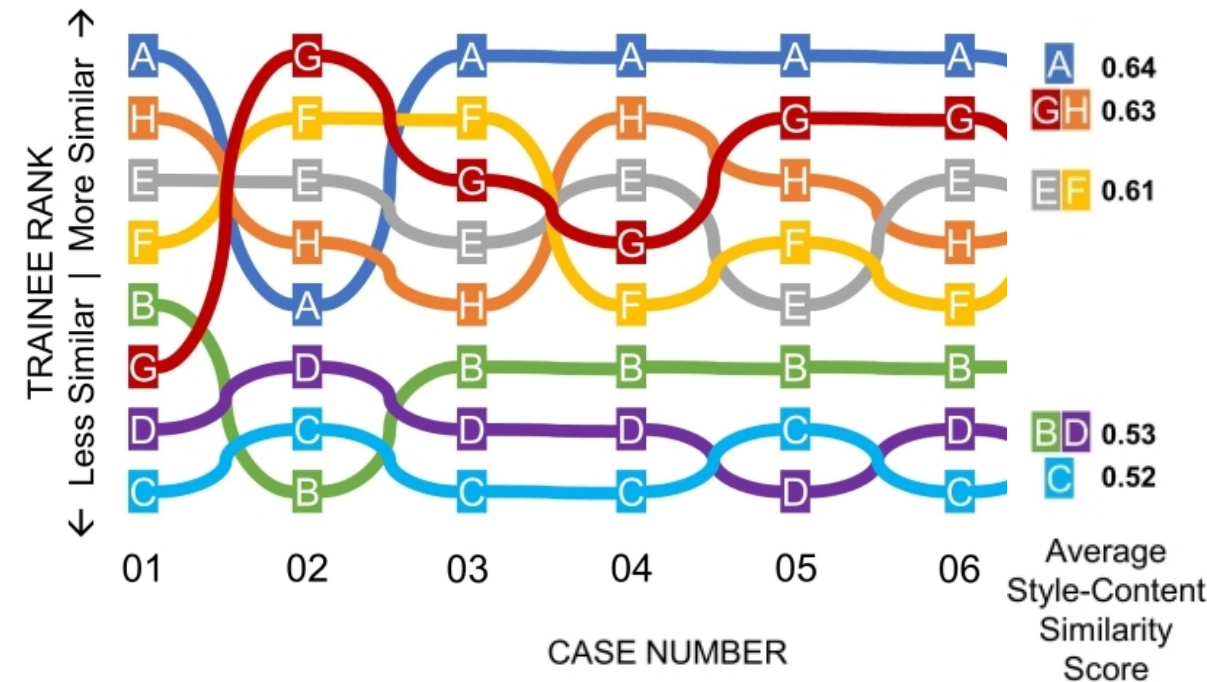
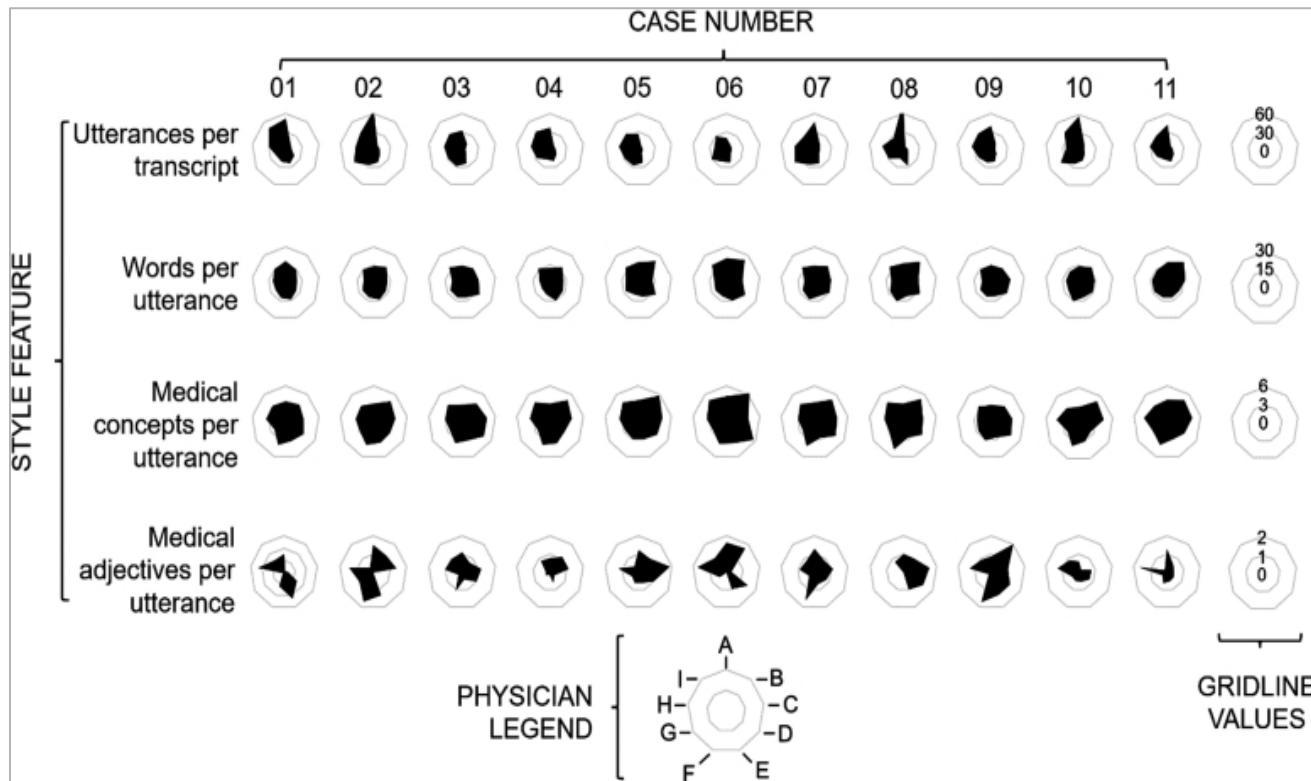
How ambient AI can revolutionize hospitalist workflow



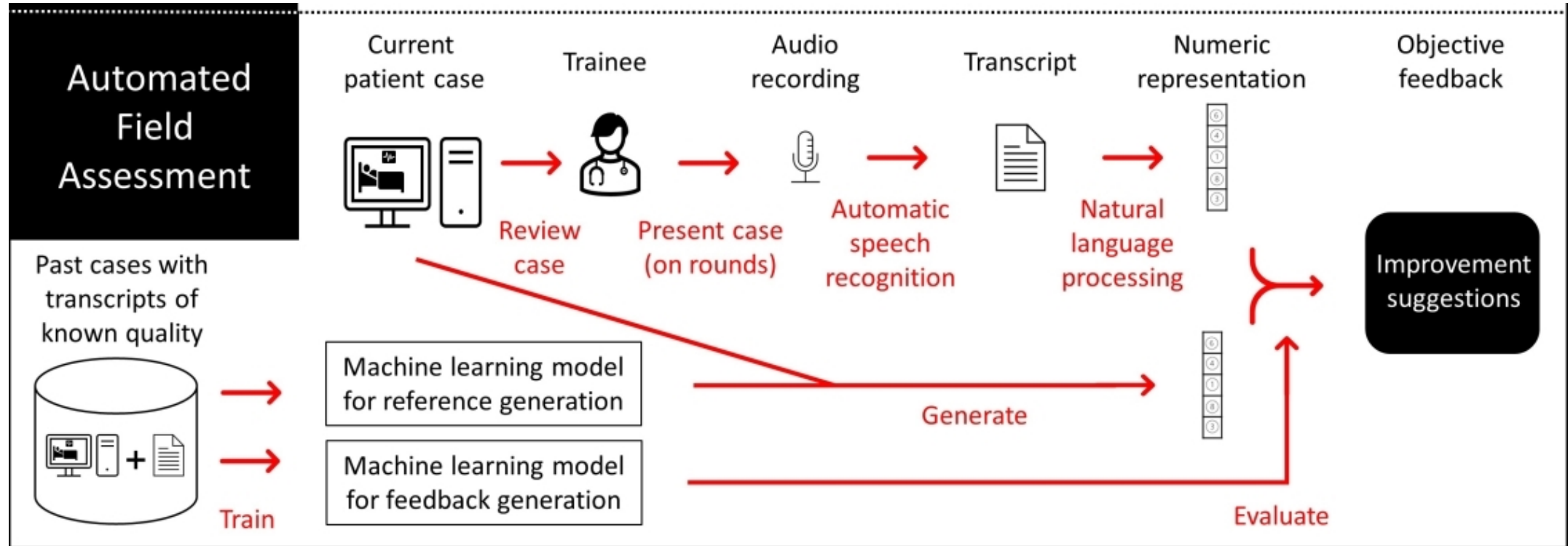
Is this Feasible? Early Work into Using Ambient Technology for Clinical Reasoning Assessments in Team Rounds



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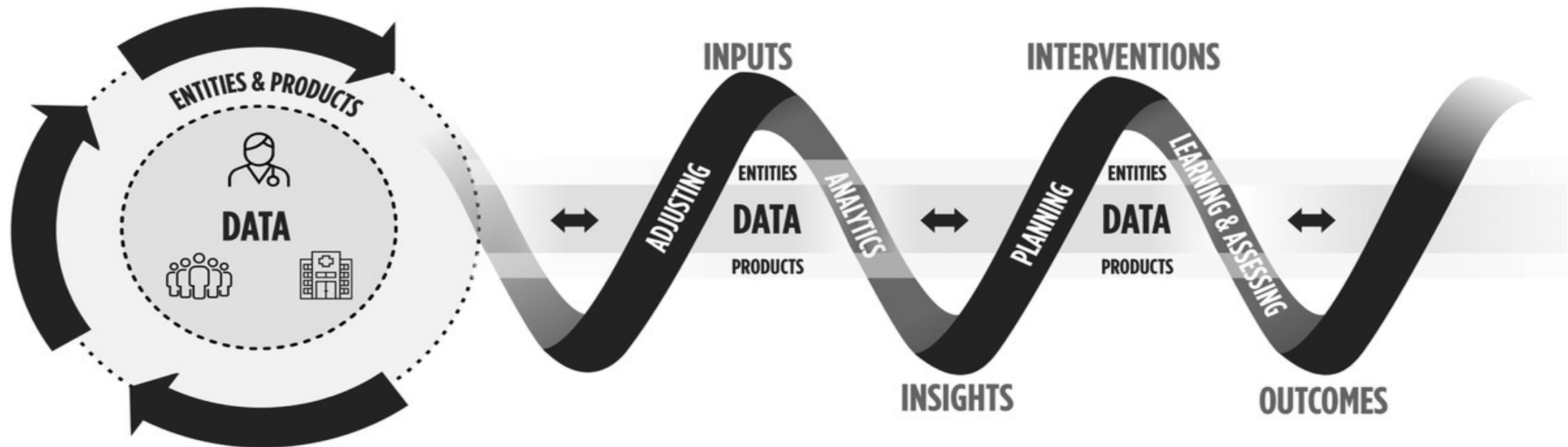
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The Future: Using Spoken Word as a Lens into Reasoning



- ▶ Measuring team-based discussions to quantify features of clinical reasoning (with semantic analysis and discourse analysis)
- ▶ Aiming to create a CR assessment tool to provide inputs/insights to help drive precision education cycle



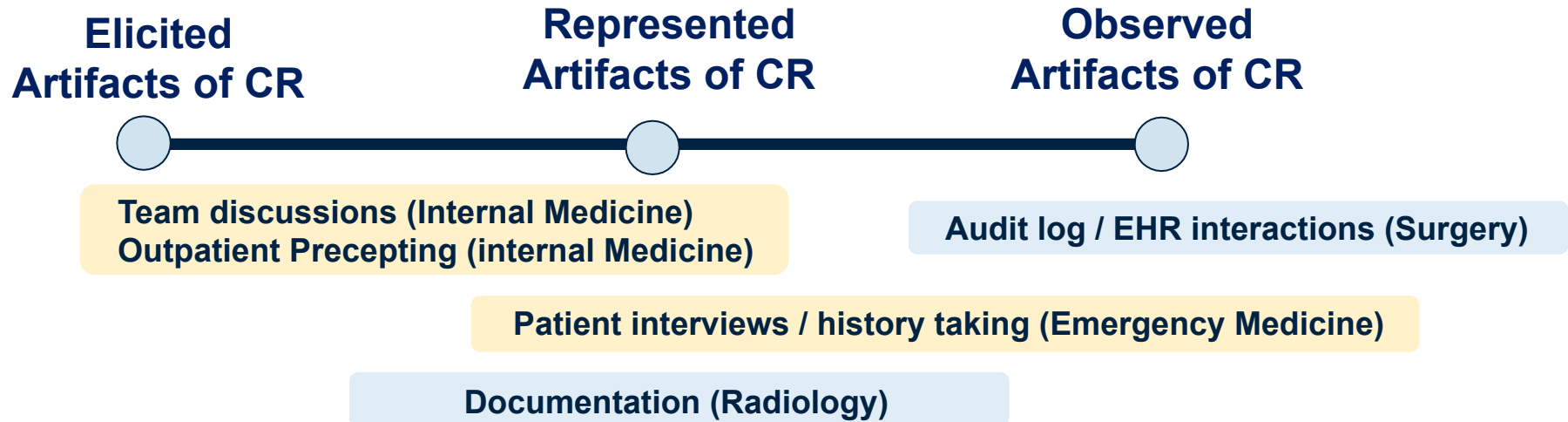
The Future: Triangulating Spoken Word With Other Artifacts



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CLINICAL REASONING INSIGHTS
FOR SHAPING PERFORMANCE

- ▶ Measuring team-based discussions to quantify features of clinical reasoning (with semantic analysis and discourse analysis)
- ▶ Aiming to create a CR assessment tool to provide inputs/insights to help drive precision education cycle
- ▶ Creating tool that captures clinical reasoning spanning artifacts across four subspecialties



Summary & Take-Home Points

- ▶ Clinical reasoning historically has been difficult to define, difficult to measure, and difficult to assess (but is paramount to clinical care)
- ▶ Ambient capture of individual presentations and team discussions offers authentic evidence of reasoning with potential for longitudinal assessment and continuing certification needs
- ▶ This novel technology provides a way to rethink traditional clinical reasoning assessments, and thus improve development of this critical skill within a precision education cycle

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- UPenn Emergency Medicine Residency
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